

SyncroTESS New Concrete Optimisation

Abstract Break Planning (Draft)

The new concrete optimisation will be enhanced to plan breaks in the realtime, preplanning and OEV mode. Due to various legal break restrictions all over the world it will be foreseen to set up the following parameters for break planning:

- n = number of breaks which should be planned during a shift.
Possible values [0,1,2]
- x_1 = length of break one in minutes
- y_1 = duration in minutes. After this duration the first break must be planned at the latest
- x_2 = length of break two in minutes
- y_2 = duration in minutes. After this duration the second break must be planned at the latest
- x_2 and y_2 only apply if $n = 2$; x_1 and y_1 apply if $n > 0$

The following rules apply:

- $n \leq 2$ breaks must be planned within a truck shift. One break of length x_1 after y_1 minutes and one break of length x_2 after y_2 minutes.
- y_1 will be respected based on the SignOn time and y_2 based on the end of break one.
- If a break takes shorter or longer than planned in reality this does not affect the length of the following break.
- Milkruns will not be interrupted by a break. If the duration of a milkrun is longer than y_1 or y_2 a break is planned directly before the milkrun.
- Breaks will not be planned during a loaded phase of a trucks turnaround but will be planned at a plant when the truck is empty.